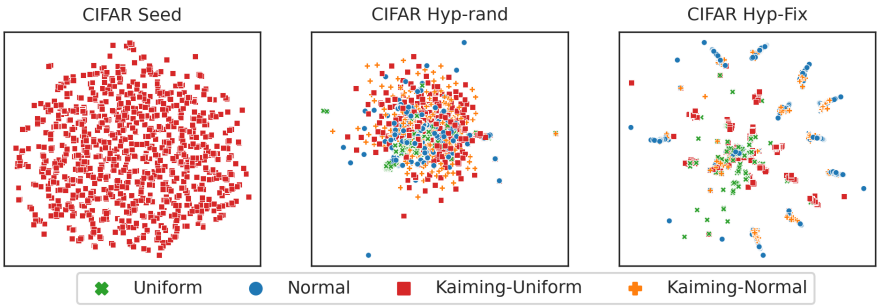

Figure 3: Visualization of the weights of the large CIFAR model zoos in different configurations. The weights are reduced to 2d using UMAP, preserving both local and global structure. In the `Seed` configuration, the UMAP reduction contains little structure. The `Hyp-rand` is equally little structured. In contrast, `Hyp-fix` contains visible clusters of initialization methods.

4 Model Zoo Analysis

The model zoos have been created aiming at diversity in generating factors, weights and performance. In this section, we analyse the zoos and their properties. Zoo cards with key values and visualizations are provided along with the zoos online. We consider models at their last epoch for the analysis. For all later analysis, non-viable checkpoints are excluded from each zoo. This includes the removal of every checkpoint with NaN values or values beyond a threshold. The threshold value is set for each zoo, such that it only excludes diverging models.

Performance To investigate the performance diversity, we consider the accuracy of the models in the zoo, see Table 2 and Figure 2. As expected, the zoos with variation only in the seed show the smallest variation in performance. Changing the hyperparameters induces a broader range of variation. Changing (Hyper-10-rand) or fixing (Hyper-10-fix) the seeds does not affect the accuracy distribution.

Model Agreement To get a more in-depth insights in the diversity of model behavior, we investigate their pairwise agreement, see Table 2. To that end, we compute the rate of agreement of class prediction between two models as $\kappa_{aggr} = \frac{1}{N}\sum_{1=1}^{N}\delta_{y_i}$. Here y_i^k, y_i^l are the predictions of models k, l for sample i of N samples. Further, $\delta_{y_i} = 1$ if $y_i^k = y_i^l$ and otherwise $\delta_{y_i} = 0$. Further, we compute the pairwise centered kernel alignment (cka) score between intermediate and last layer outputs and denote it as κ_{cka}. The cka score evaluates the correlation of activations, compensating for equivariances typical for neural networks [44]. In empirical evaluations, we found the cka score robust for relatively small number of image samples, and compute the scrore using 50 images to reduce the computational load. Both agreement metrics confirm the expectation and performance results. Zoos with higher overall performance naturally have a higher agreement on average, as there fewer samples on which to disagree. Zoos with varying hyperparameters(Hyp-10-rand and Hyp-10-fix) agree less on average than zoos with changes in seed only (Seed). What is more, the distribution of κ_{aggr} and κ_{cka} in the `Seed` zoos is unimodal and approximately gaussian. In the `Hyp-10` zoos, the distributions are bi-modal, with one mode around 0.1 (0.0) and the other around 0.9 (0.75) in hard agreement (cka score). In these zoos, models agree to a rather high degree with some models, and disagree with others.

Weights Lastly, we investigate the diversity of the model zoos in weight space, see again Table 2. By design, the mean weight value of the zoos varying only in the seed is larger than in the other zoos, while the standard deviation does not differ greatly (Table 2, column w). To get a better intuition in the distribution of models in weight space, we compute the pairwise $\ell_2(\mathbf{w_k}, \mathbf{w_l}) = \frac{\|\mathbf{w_k}-\mathbf{w_l}\|_2^2}{1/N\sum_{n=1}^{N}\|\mathbf{w_n}\|_2^2}$ and cosine distance $cos(\mathbf{w_k}, \mathbf{w_l}) = 1 - \frac{\mathbf{w_l}^T\mathbf{w_k}}{\|\mathbf{w_k}\|_2^2\|\mathbf{w_l}\|_2^2}$, and investigate their distribution. Here, too, varying the hyperparameters introduces higher amounts of diversity, while changing or fixing the seeds does not affect the weight diversity much. As these values are computed at the end of model training, repeated starting points due to fixed seeds appear not to reduce weight diversity significantly. In a more hands-off approach, we compute 2d reductions of the weight over all epochs using UMAP [40]. In the 2d reductions (see Figure 3), the zoos varying in seed only show little to no structure.

Table 3: Benchmark results for predicting model properties from the weights ($\mathbf{w}$) and layer-wise weight statistics ($s(\mathbf{w})$) using linear models. We report the prediction R^2 for accuracy, generalization gap (GGap), epoch, learning rate (LR) and dropout (Drop) and prediction accuracy for initalization method (Init) and activation function (Act). Values reported in %, higher values are better.

			Accuracy		GGap		Epoch		Init		Act	
Dataset	Architecture	Config	$\mathbf{w}$	$s(\mathbf{w})$	$\mathbf{w}$	$s(\mathbf{w})$	$\mathbf{w}$	$s(\mathbf{w})$	$\mathbf{w}$	$s(\mathbf{w})$	$\mathbf{w}$	$s(\mathbf{w})$
MNIST	CNN (s)	Seed	92.3	98.7	2.1	68.8	87.2	97.8	n/a	n/a	n/a	n/a
	CNN (s)	Hyp-10-r	-11.2	69.4	-49.8	13.7	-95.5	14.3	42.6	77.6	45.5	78.5
	CNN (s)	Hyp-10-f	66.5	70.1	5.4	12.5	-4.8	14.5	94.3	79.8	81.2	76.8
F-MNIST	CNN (s)	Seed	87.5	97.2	20.9	60.5	89.1	97.1	n/a	n/a	n/a	n/a
	CNN (s)	Hyp-10-r	8.7	76.9	-47.5	13.7	-70.1	18.9	48.4	81.5	47.9	79.6
	CNN (s)	Hyp-10-f	62.4	75.6	3.9	12.6	-2.0	17.0	95.4	81.6	84.6	77.7
SVHN	CNN (s)	Seed	91.0	98.6	-42.8	65.9	66.9	92.5	n/a	n/a	n/a	n/a
	CNN (s)	Hyp-10-r	-8.6	90.3	-55.3	27.6	-30.5	11.1	38.2	58.5	55.7	72.3
	CNN (s)	Hyp-10-f	64.2	89.9	17.5	27.4	-0.1	11.1	67.3	58.2	76.1	73.6
USPS	CNN (s)	Seed	92.5	98.7	44.3	71.8	86.0	98.4	n/a	n/a	n/a	n/a
	CNN (s)	Hyp-10-r	-11.5	70.3	-35.2	13.6	-75.7	21.3	49.2	88.8	43.7	66.2
	CNN (s)	Hyp-10-f	73.2	70.8	10.8	14.7	18.9	23.0	96.3	88.1	74.5	72.7
CIFAR10	CNN (s)	Seed	75.3	96.0	27.0	90.2	68.6	91.1	n/a	n/a	n/a	n/a
	CNN (s)	Hyp-10-r	50.1	88.0	-4.3	40.5	-2.7	34.2	34.0	50.5	71.5	80.9
	CNN (s)	Hyp-10-f	67.0	87.9	38.2	42.9	27.0	31.8	72.0	52.2	75.6	80.0
CIFAR10	CNN (l)	Seed	83.6	98.2	33.4	92.9	86.5	95.7	n/a	n/a	n/a	n/a
	CNN (l)	Hyp-10-r	32.6	90.5	-0.9	47	-10.5	35.5	41.6	51.6	69.1	83.1
	CNN (l)	Hyp-10-f	64.5	91.4	30.4	40.7	29.8	35.3	74.5	54.9	77.7	86.0
STL	CNN (s)	Seed	17.8	91.2	2.0	30.2	45.3	95.0	n/a	n/a	n/a	n/a
	CNN (s)	Hyp-10-r	-8.7	77.1	-44.0	9.3	-68.8	19.1	41.3	93.9	46.3	66.8
	CNN (s)	Hyp-10-f	76.1	76.5	6.7	10.7	21.2	22.4	98.1	91.3	78.1	62.6
STL	CNN (l)	Seed	-112	94.2	2.8	37.3	5.6	98.7	n/a	n/a	n/a	n/a
	CNN (l)	Hyp-10-r	-79.6	74.1	-118	10.7	-106	18.8	43.8	90.4	49.4	68.3
	CNN (l)	Hyp-10-f	84.1	77.7	10.4	11.7	14.6	19.1	97.8	92.8	78.8	68.0
CIFAR10	ResNet-18	Seed	–.-	96.8	–.-	76.7	–.-	99.6	n/a	n/a	n/a	n/a
CIFAR100	ResNet-18	Seed	–.-	97.4	–.-	95.4	–.-	99.9	n/a	n/a	n/a	n/a
t-ImageNet	ResNet-18	Seed	–.-	96.1	–.-	87.5	–.-	99.9	n/a	n/a	n/a	n/a

Zoos with hyperparameter changes and random seeds are similarly unstructured. Zoos with varying hyperparameters and fixed seeds show clear clusters with models of the same initialization method and activation function. These findings are further supported by the predictability of initialization method and activation function (Table 3). The structures are unsurprising considering that the activation function is very influential in shaping the loss surface, while initialization method and the seed determine the starting point on it. Depending on the downstream task, this property can be desirable or should be avoided, which is why we provide both configurations.

Model Property Prediction As a set of benchmark results on the proposed model zoos and to further evaluate the zoos, we use linear models to predict hyperparameters or performance values of the individual models. As features, we use the model weights $\mathbf{w}$ or per-layer quintiles of the weights $s(\mathbf{w})$ as in [55]. Linear models are used to evaluate the properties of the dataset and the quality of the features. We report these results in Table 3. The layer-wise weight statistics ($s(\mathbf{w})$) have generally higher predictive performance than the raw weights $\mathbf{w}$. In particular, $s(\mathbf{w})$ are not affected by using fixed or random seeds and thus generalize well to unseen seeds. For the ResNet-18 zoos, $\mathbf{w}$ becomes too large to be used as a feature and is therefore omitted. Across all zoos, the accuracy as well as the hyperparameters can be predicted very accurately. Generalization gap and epoch appear to be more difficult to predict. These findings hold for all zoos, regardless of the different architectures, model sizes, task complexity and performance range. $\mathbf{w}$ can be used to predict the initialization method and activation function to very high accuracy, if the seeds are fixed. The performance drops drastically if seeds are varied. This results confirms our expectation of diversity in weight space induced by fixing or varying seed. These results show i) that the model weights of our zoos contain rich information on their properties; ii) confirm the notions of diversity that were design goals for the zoos; and iii) leave room for improvements on the more difficult properties to predict, in particular the generalization gap.

5 Potential Use-Cases & Applications

While populations of NNs have been used in previous work, they still are relatively novel as a dataset. As use-cases for such datasets may not be obvious, this section presents potential use-cases and applications. For all use-cases, we collect related work that uses model populations. Here, the zoos may be used as data or to evaluate the methods. For some of the use-cases, the analysis above provides support. Lastly, we suggest ideas for future work which we hope can inspire the community to make use of the model zoos.

5.1 Model Analysis

The analysis of trained models is an important and difficult step in the machine learning pipeline. Commonly, models are applied on hold-out test sets, which may contain difficult cases with specific properties [31]. Other approaches identify subsections of input data that is relevant for a specific output [61, 25, 66]. A third group of methods compares the activations of models, e.g. the cka method used in Sec. 4 to measure diversity [27].

Populations of models have been used to identify commonalities in model weights, activations, or graph structure which are predictive for model properties. Some methods use the weights, weight-statistics or eigenvalues of the weight matrices as features to predict a model's accuracy or hyper-parameters [55, 12, 39]. Recently, [50] have learned self-supervised representation of the weights and demonstrate their usefulness for predicting model properties. Other publications use activations to approximate intermediate margins [58, 24] or graph connectivity features [6] to predict the generalization gap or test accuracy. Standardized, diverse model zoos may facilitate development of new methods, or be used as evaluation dataset for existing model analysis, interpretability or comparison method.

Previous work as well as the experiment results in Sec 4 indicate that even more complex model properties might be predicted from the weights. By studying populations of models, in-depth diagnostics of models, such as whether a model learned a specific bias, may be based on the weights or topology of models. Lastly, model properties as well of the weights may be used to derive a model 'identity' along the training trajectory, to allow for NN versioning.

5.2 Learning Dynamics

Analysing and utilizing the learning dynamics of models has been a useful practice. For example, early stopping [15], which determines when to end training at minimal generalization error based on a cross validation set and has become standard in machine learning practice.

More recently, methods have exploited zoos of models. Population based training [23] evaluates the performance of model candidates in a population, decides which of the candidates to pursue further and which to give up. HyperBand evaluates performance metrics for groups of models to optimize hyperparameters [34, 33]. Research in Neural Architecture Search was greatly simplified by the NASBench dataset family [59], which contains performance metrics for varying hyperparameter choices. Our model zoos extend these datasets by adding models including their weights at states throughout training, which may open new doors for new approaches.

The accuracy distribution of our model zoos become relatively broad if hyperparameters are varied (Figure 2). For early stopping or population based methods, identifying a good range of hyperparameters to try, and then identifying those candidates that will perform best towards the end of training, is a challenging and relevant task. Our model zoos may be used to develop and evaluate methods to that end. Beyond that, diverse model zoos offer the opportunity to make further steps of understanding and exploiting the learning dynamics of models, i.e., by studying the regularities of generalizing and overfitting models. The shape and curvature of training trajectories may contain rich information on the state of model training. Such information could be used to monitor model training, or adjust hyperparameters to achieve better results. The sparsified model zoos add several potential use-cases. They may be used to study the sparsification performance on a population level, study emerging patterns of populations of sparse models, or the relation of full models and their sparse counterparts.